

Human-Computer Interaction

Capturing and utilizing human silhouette in interface design (2)

- This set of exercises extends the previous class on capturing and utilizing the human silhouette.
- The aim is to explore the use of hand motion and gestures as an interface for human-computer interaction.
- The tasks demonstrate how to translate hand landmarks detected with the MediaPipe library into cursor movements and simple in-air actions.

Exercise 1 – Translate the movement of the index finger into cursor movement on the screen

- Use pyautogui library
 - `import pyautogui`
- Read screen dimensions
 - `screen_width, screen_height = pyautogui.size()`
- Find the landmark for the index finger (it's index no 8)
 - `index_finger_landmark = hand_landmarks.landmark[8]`
- Convert the landmark position to mouse coordinates
 - `x, y = int(index_finger_landmark.x * screen_width),
int(index_finger_landmark.y * screen_height)`
- Move the mouse to the calculated coordinates
 - `pyautogui.moveTo(x, y, duration=0.1)`

Exercise 2 - Suggest a mechanism for handling in-air clicks

- Based on the experience gained, propose a mechanism that allows left-clicking with the mouse
- Implement and integrate it into the code developed so far
- Think about how you could distinguish a click gesture from normal cursor movement.
- You may consider:

- monitoring the relative distance between selected landmarks (e.g., the tips of the index finger and the thumb) — when the distance becomes small enough, it can be interpreted as a click gesture;
- using a “hover to select” mechanism, where keeping the cursor steady over an object for a certain period of time triggers a selection or click action.
- Once you detect the desired condition for a click, you can trigger a mouse action using the PyAutoGUI library, for example:
 - `pyautogui.click()` # performs a left-click
 - `pyautogui.doubleClick()` # performs a double-click
 - `pyautogui.rightClick()` # performs a right-click

Exercise 3 - Implement right-click functionality

- Extend your existing code to allow triggering the right mouse button.
- Consider which hand landmarks or gesture patterns could reliably differentiate a right-click from a left-click.
- Test your solution and discuss its reliability with the group – think about how intuitive and robust the gesture is.

Exercise 4 – Implementation of in-air push gesture

- Examine the third coordinate returned in the landmarks for depth.
- Remember that the *Z* coordinate is estimated and may be subject to a larger error than the other coordinates.
- Utilizing the chosen landmark on the hand or the average value of several landmarks, develop the code to detect pressing in the air.
- Evaluate your proposal.

Report

- Prepare a short report summarizing your findings and reflections. Include a discussion of the results and attach the code you developed.
- Send your report to: anowosielski@zut.edu.pl
 - In the e-mail title, please use the scheme: HCI Your Full Name